

MERCER COUNTY AP, WV, USAWMO: **724125**Lat: **37.298N**Lon: **81.204W**Elev: **871**StdP: **91.30**Time Zone: **-5.00 (NAE)**Period: **94-19**WBAN: **03859****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.4	-10.8	-20.2	0.7	-9.1	-17.8	0.9	-6.9	10.7	4.2	9.4	3.6	3.8	300	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	29.9	20.5	28.6	19.9	27.6	19.5	22.0	27.2	21.4	26.4	20.8	25.6	2.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	16.8	24.5	19.8	16.2	23.8	19.3	15.6	23.3	68.7	27.4	66.3	26.5	64.1	25.6	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.8	7.0	-16.6	31.3	3.1	1.9	-18.9	32.7	-20.7	33.8	-22.5	34.9	-24.8	36.3		
(5)			-17.2	23.1	3.0	0.7	-19.4	23.6	-21.1	24.0	-22.8	24.5	-25.0	25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	0.8	2.4	6.8	13.0	17.1	20.8	22.3	22.0	18.9	13.1	7.5	3.0
	DBStd	9.16	6.93	6.17	6.37	5.56	4.41	2.88	2.37	2.47	3.50	5.02	5.58	6.23
	HDD10.0	1064	294	221	139	34	4	0	0	0	1	28	113	229
	HDD18.3	2616	544	445	358	170	76	12	2	3	33	171	327	474
	CDD10.0	1928	8	9	41	125	224	324	382	371	269	126	36	13
	CDD18.3	439	0	0	2	11	38	86	126	116	51	9	1	0
	CDH23.3	2322	0	0	6	69	199	462	673	611	262	37	2	0
	CDH26.7	449	0	0	0	7	27	90	148	125	48	4	0	0
Wind	WSAvg	3.0	3.9	3.9	3.7	3.6	2.8	2.4	2.2	1.9	2.1	2.8	3.1	3.5
Precipitation	PrecAvg	980	67	68	87	88	105	89	109	82	81	65	65	73
	PrecMax	1303	138	157	209	220	202	197	260	141	195	152	143	123
	PrecMin	764	18	15	38	12	36	19	47	23	10	2	12	7
	PrecStd	133	28	30	37	40	37	43	44	29	43	36	32	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.3	19.0	24.0	27.6	29.1	30.8	31.4	31.3	30.1	26.5	22.1	19.3
		MCWB	12.8	12.4	14.3	14.8	18.1	20.6	21.5	21.3	18.7	17.3	13.6	13.6
	2%	DB	15.4	16.4	20.8	25.3	27.2	28.9	29.7	29.4	28.1	24.2	19.6	16.7
		MCWB	10.8	10.9	12.5	14.5	17.6	20.2	21.1	20.7	18.8	15.9	12.2	12.5
	5%	DB	13.3	13.9	18.2	23.4	25.6	27.5	28.3	28.1	26.3	22.2	17.5	14.6
		MCWB	9.8	9.1	11.2	13.6	17.2	19.4	20.6	20.4	18.4	15.3	11.7	10.5
	10%	DB	11.0	11.7	16.1	21.4	23.9	26.1	27.0	26.8	24.5	20.3	15.7	12.4
		MCWB	7.4	7.4	10.3	12.9	16.6	19.1	20.2	20.0	17.8	14.8	10.8	9.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.6	14.0	15.1	17.1	20.1	22.1	23.2	22.9	21.3	19.7	15.6	14.9
		MCDB	17.0	17.7	22.1	24.0	26.2	28.1	28.5	28.6	26.4	22.6	19.7	18.2
	2%	WB	12.0	12.4	13.7	16.0	19.1	21.2	22.3	21.9	20.5	18.3	14.2	13.0
		MCDB	15.1	15.7	19.0	22.1	24.5	27.0	27.4	26.9	24.9	21.7	17.6	15.9
	5%	WB	10.2	10.1	12.5	15.1	18.3	20.5	21.6	21.3	19.7	17.0	13.0	11.3
		MCDB	12.8	12.8	17.1	20.9	23.5	25.9	26.5	26.1	23.7	20.8	16.1	14.1
	10%	WB	7.9	8.0	11.2	14.0	17.5	19.9	20.9	20.7	19.0	15.6	11.4	9.3
		MCDB	10.4	11.1	15.1	19.5	22.5	24.5	25.6	25.2	22.9	19.2	14.9	11.9
Mean Daily Temperature Range	5% DB	MDBR	8.4	8.8	9.5	10.3	9.4	9.1	8.5	8.6	9.0	9.0	9.1	7.9
		MCDBR	9.6	10.8	11.1	11.9	10.9	10.4	10.0	9.9	10.9	10.8	10.2	9.5
		MCWBR	7.1	7.1	6.1	5.3	4.5	4.4	4.0	3.8	4.3	4.9	5.9	6.6
	5% WB	MCDBR	8.8	9.2	9.9	9.9	9.4	9.2	8.8	8.5	8.6	8.7	8.4	9.3
		MCWBR	7.3	7.6	6.3	4.9	4.4	4.3	3.9	3.7	3.9	4.6	6.4	7.4
Clear-Sky Solar Irradiance	taub	0.287	0.295	0.318	0.352	0.391	0.415	0.443	0.452	0.354	0.330	0.299	0.282	
	taud	2.524	2.487	2.475	2.399	2.278	2.246	2.194	2.146	2.417	2.492	2.551	2.562	
	Ebn at Noon	912	945	949	928	891	866	838	822	896	891	890	896	
	Edh at Noon	77	91	102	116	133	138	144	148	107	90	75	70	
All-Sky Solar Radiation	RadAvg	2.09	2.75	3.85	4.97	5.51	6.02	5.63	5.23	4.43	3.41	2.52	1.83	
	RadStd	0.20	0.35	0.28	0.39	0.44	0.41	0.35	0.26	0.42	0.42	0.27	0.19	